Web Technologies (CSC336) - Lab Mid Term

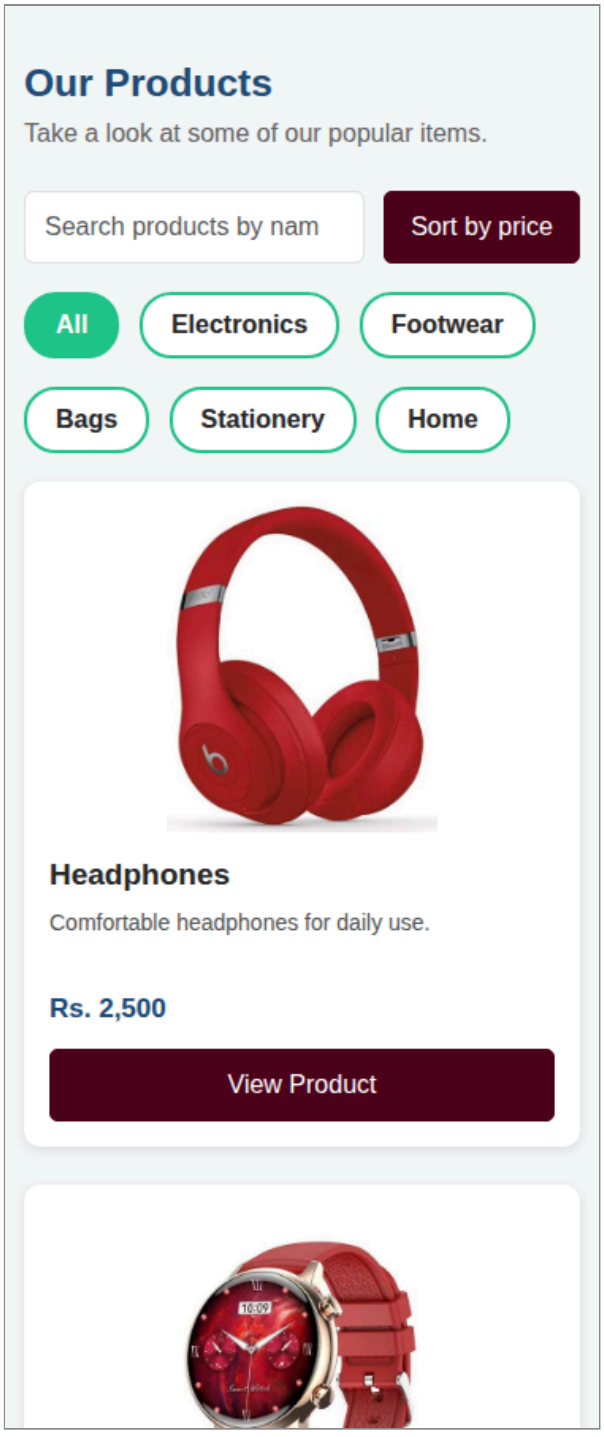
Name: Your Name **Registration No.:** FA26-BCS-074

COMSATS University Islamabad, Lahore Campus - FA26 BCS A

N	BP = 600 + N	H = (N x 7) mod 360	K = (N mod 3) + 3	Sticky top
74	674px	158	5	104px

Task 1 - CSS-only category filter

Screenshots (375px, 674px, 1280px)



Task 1 - 375px - page loaded, "All" selected

Our Products

Take a look at some of our popular items.

Search products by nam

Sort by price

All | Electronics | Footwear | Bags | Stationery | Home



Travel Bag

A useful bag for university and travel.

Rs. 1,800

View Product



Task 1 - 375px - "Bags" selected (only matching cards, no gaps)

Our Products

Take a look at some of our popular items.

Search products by name...

Sort by price

All | Electronics | Footwear | Bags | Stationery | Home



Headphones

Comfortable headphones for daily use.

Rs. 2,500

View Product



Smart Watch

A simple watch for everyday use.

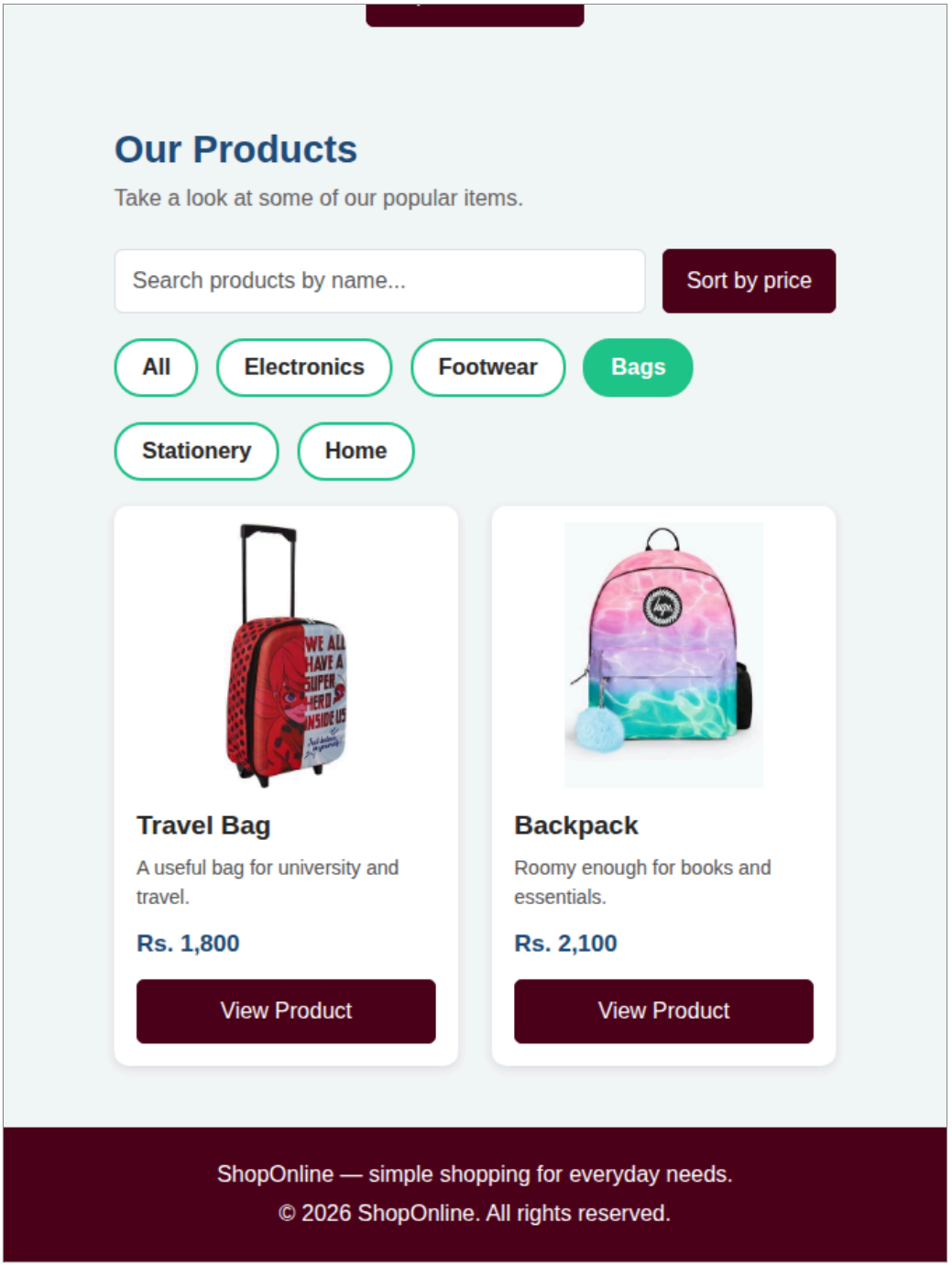
Rs. 4,000

View Product





Task 1 - 674px - page loaded, "All" selected



Task 1 - 674px - "Bags" selected (only matching cards, no gaps)

Our Products

Take a look at some of our popular items.

Search products by name...

Sort by price

All | Electronics | Footwear | Bags | Stationery | Home



Headphones

Comfortable headphones for daily use.

Rs. 2,500

View Product



Smart Watch

A simple watch for everyday use.

Rs. 4,000

View Product



Sports Shoes

Lightweight shoes for daily comfort.

Rs. 3,200

View Product



Travel Bag

A useful bag for university and travel.

Rs. 1,800

View Product









Task 1 - 1280px - page loaded, "All" selected

Explore Products

Our Products

Take a look at some of our popular items.

Search products by name...

Sort by price

All | Electronics | Footwear | Bags | Stationery | Home



Travel Bag

A useful bag for university and travel.

Rs. 1,800

View Product



Backpack

Roomy enough for books and essentials.

Rs. 2,100

View Product

ShopOnline — simple shopping for everyday needs.

© 2026 ShopOnline. All rights reserved.

Task 1 - 1280px - "Bags" selected (only matching cards, no gaps)

Code added / changed

index.html

```
<!-- TASK 1 START: CSS-only category filter (radio buttons must be siblings of the grid) -->
            <div class="filter-wrap">
                <input type="radio" name="category" id="cat-all" value="all" checked>
                <label for="cat-all">All</label>
                <input type="radio" name="category" id="cat-electronics" value="electronics">
                <label for="cat-electronics">Electronics</label>
                <input type="radio" name="category" id="cat-footwear" value="footwear">
                <label for="cat-footwear">Footwear</label>
                <input type="radio" name="category" id="cat-bags" value="bags">
                <label for="cat-bags">Bags</label>
                <input type="radio" name="category" id="cat-stationery" value="stationery">
                <label for="cat-stationery">Stationery</label>
                <input type="radio" name="category" id="cat-home" value="home">
                <label for="cat-home">Home</label>

                <div class="row g-4" id="product-grid">

    <!-- every product card column got 'product-item' and a data-category: -->
                <div class="col-12 col-sm-6 col-lg-3 product-item" data-category="electronics">
                <div class="col-12 col-sm-6 col-lg-3 product-item" data-category="electronics">
                <div class="col-12 col-sm-6 col-lg-3 product-item" data-category="footwear">
                <div class="col-12 col-sm-6 col-lg-3 product-item" data-category="bags">
                <div class="col-12 col-sm-6 col-lg-3 product-item" data-category="bags">
                <div class="col-12 col-sm-6 col-lg-3 product-item" data-category="home">
                <div class="col-12 col-sm-6 col-lg-3 product-item" data-category="stationery">
                <div class="col-12 col-sm-6 col-lg-3 product-item" data-category="home">
    ... (cards themselves are unchanged) ...

                </div>  <!-- end of #product-grid -->
            </div>  <!-- end of .filter-wrap -->
            <!-- TASK 1 END -->
```

style.css

```
/* ===== TASK 1 START: CSS-only category filter ===== */
/* Accent colour H = 158: hsl(158, 70%, 45%) */
:root {
    --accent: hsl(158, 70%, 45%);
}

/* the real radio buttons are hidden; the labels are the visible buttons */
.filter-wrap {
    position: relative;
}

.filter-wrap input[type="radio"] {
    position: absolute;
    opacity: 0;
}

.filter-wrap label {
    display: inline-block;
    padding: 7px 18px;
    margin: 0 8px 18px 0;
    border: 2px solid var(--accent);
    border-radius: 20px;
    color: #292929;
    background: white;
    font-weight: bold;
    cursor: pointer;
}

/* the selected filter looks selected (uses the accent colour) */
.filter-wrap input[type="radio"]:checked + label {
    background: var(--accent);
    color: white;
}

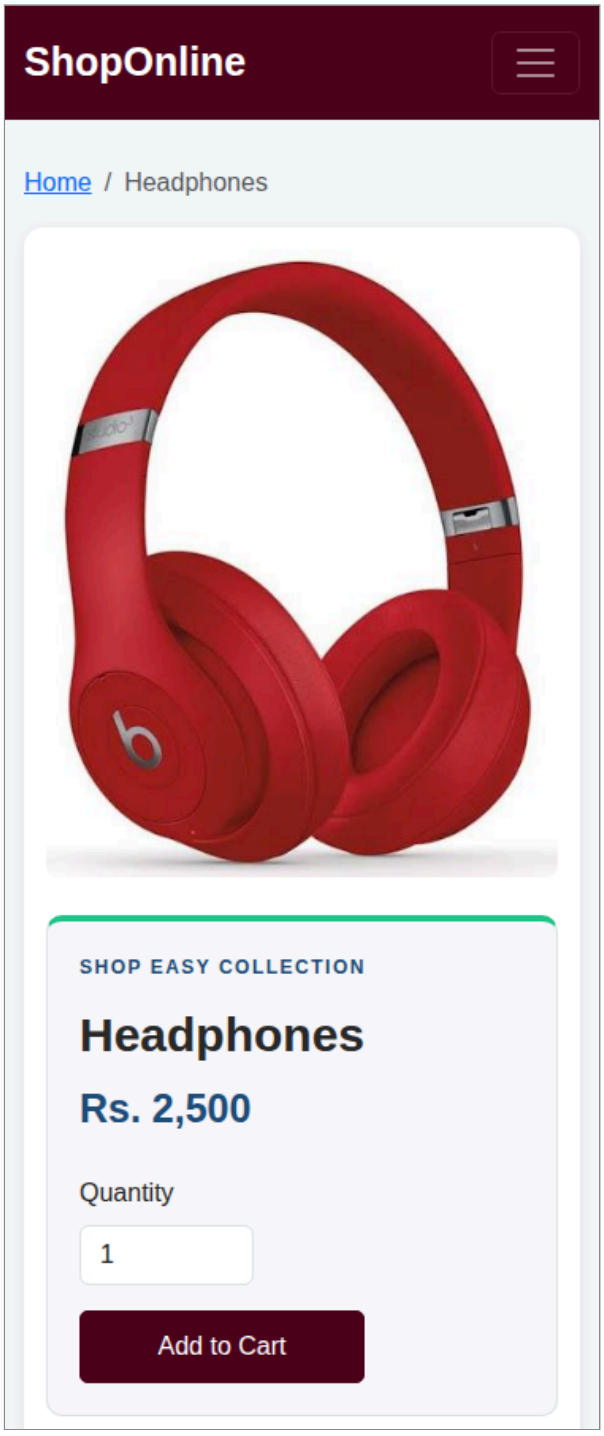
/* keyboard focus outline */
.filter-wrap input[type="radio"]:focus-visible + label {
    outline: 3px solid #292929;
    outline-offset: 2px;
}

/* When a category radio is checked, hide every card whose data-category is different.
   display:none removes the card from the layout, so there are no empty gaps. */
#cat-electronics:checked ~ .row .product-item:not([data-category="electronics"]) { display: none; }
```

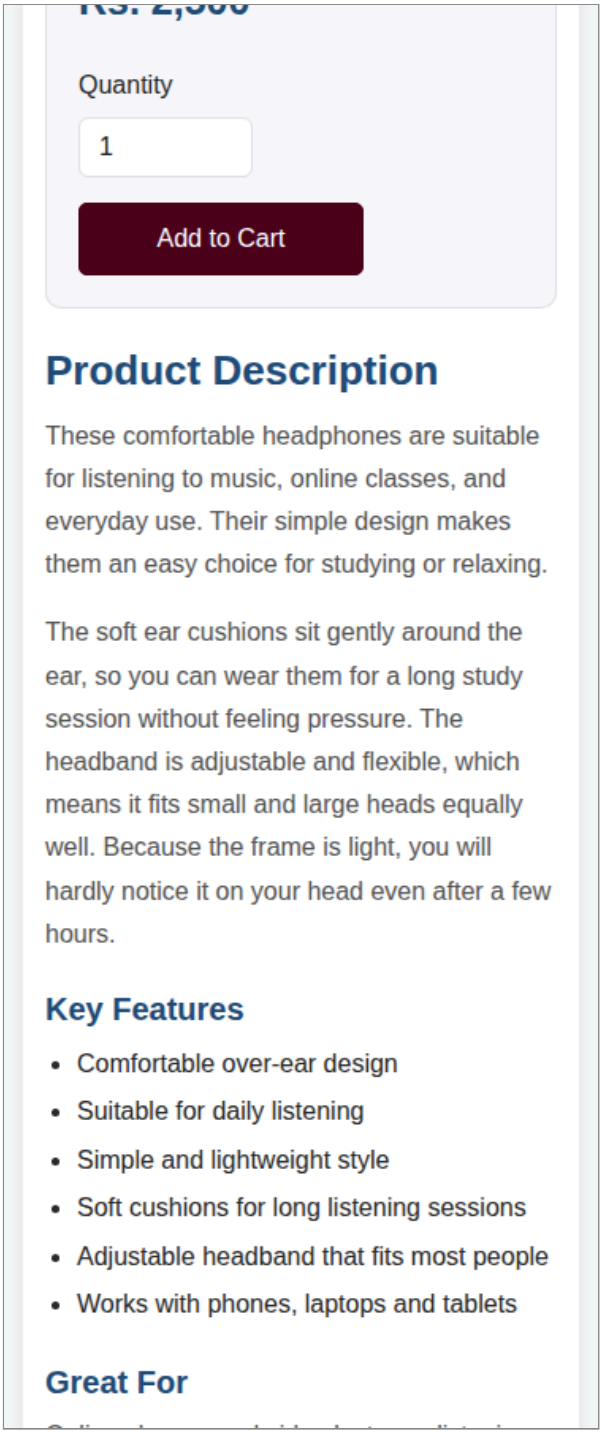
```
#cat-footwear:checked ~ .row .product-item:not([data-category="footwear"]) { display: none; }
#cat-bags:checked ~ .row .product-item:not([data-category="bags"]) { display: none; }
#cat-stationery:checked ~ .row .product-item:not([data-category="stationery"]) { display: none; }
#cat-home:checked ~ .row .product-item:not([data-category="home"]) { display: none; }
/* ===== TASK 1 END ===== */
```

Task 2 - Product page layout with a sticky details box

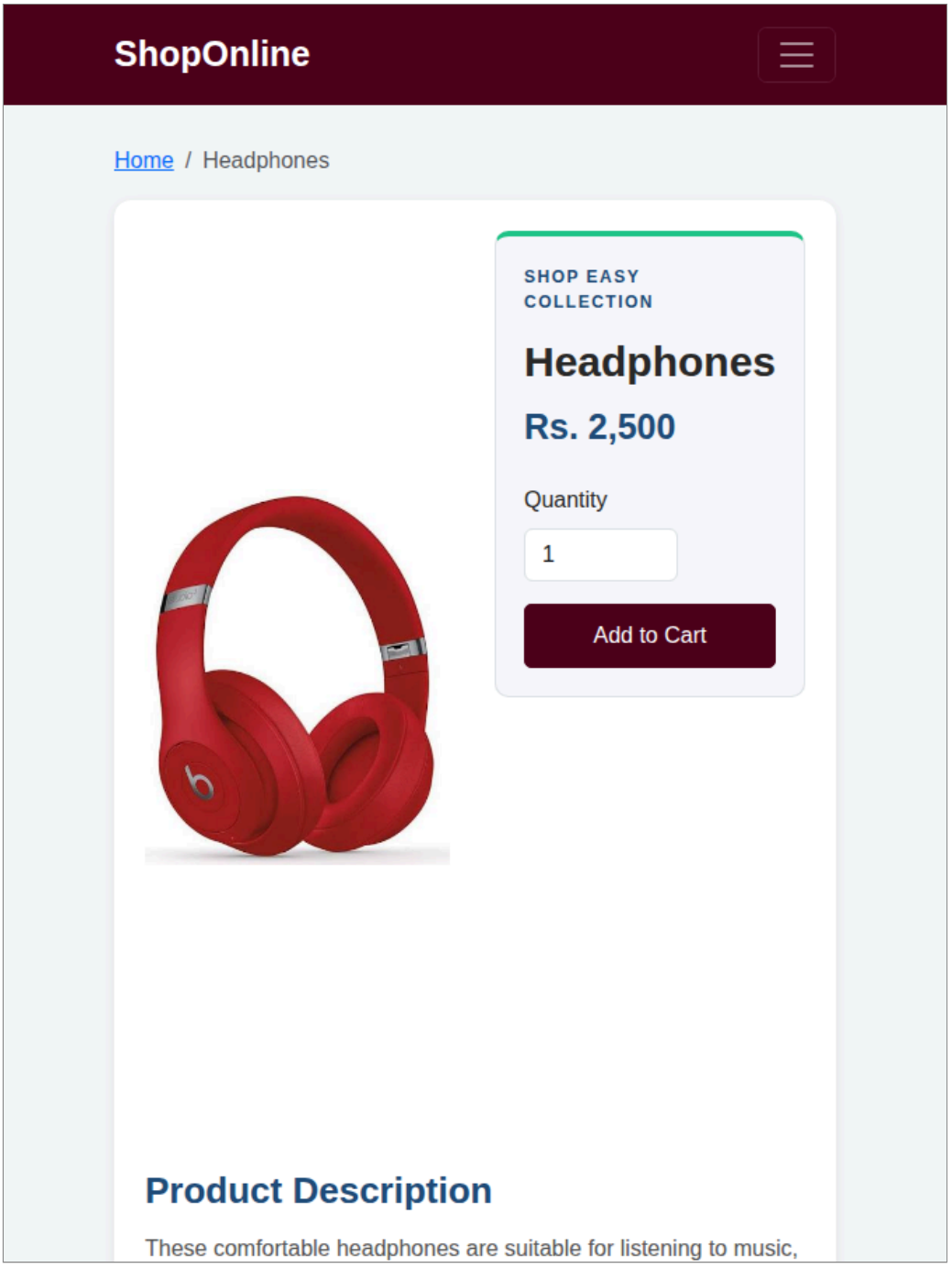
Screenshots (375px, 674px, 1280px)



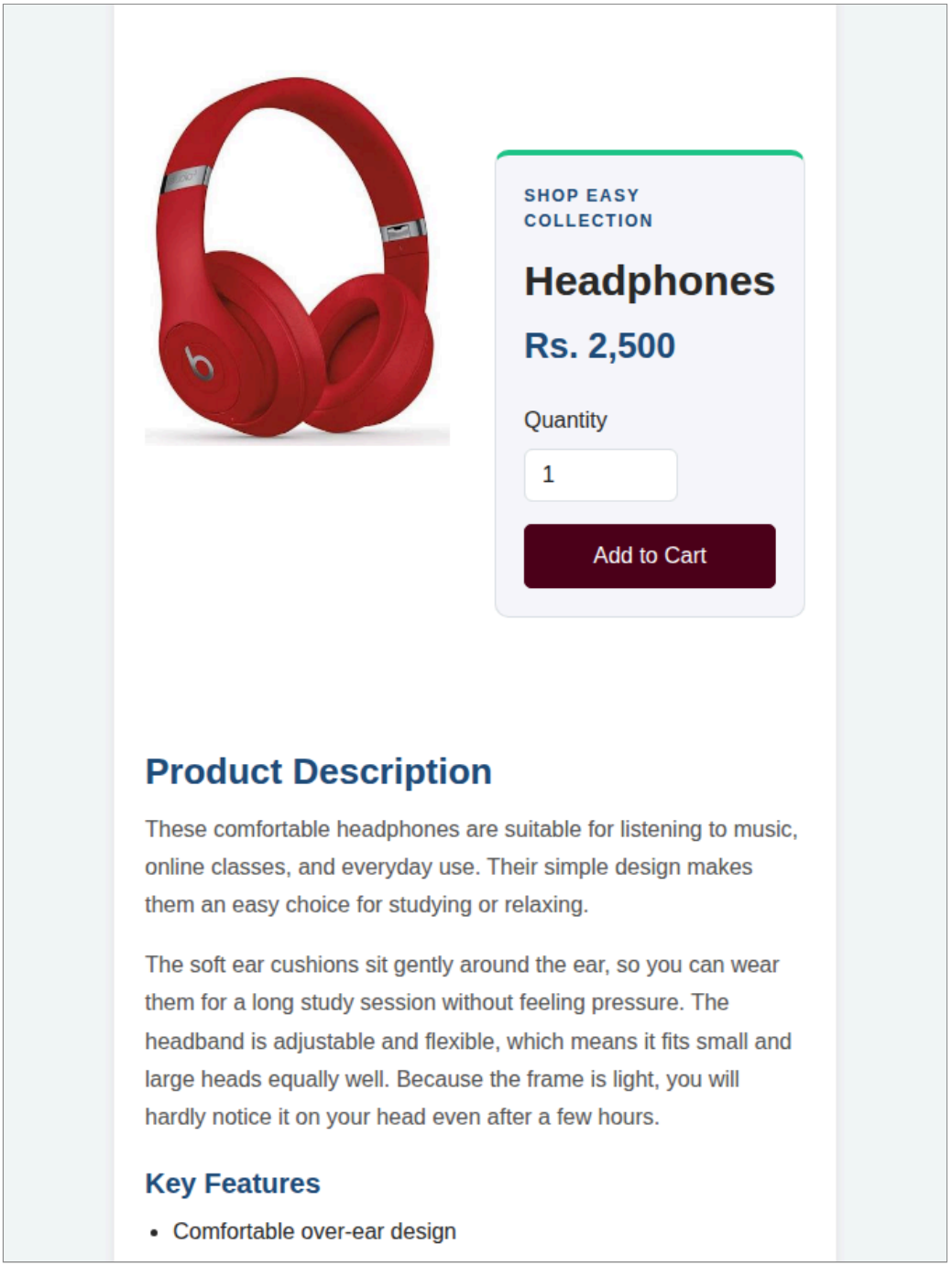
Task 2 - 375px - top of the product page



Task 2 - 375px - scrolled down: details, then description (stacked, not sticky)



Task 2 - 674px - top of the product page



Task 2 - 674px - scrolled down: details box is sticky, 104px from the top

ShopOnline Home Products About Contact Cart

Home / Headphones



SHOP EASY COLLECTION

Headphones

Rs. 2,500

Quantity

1

Add to Cart

Product Description

These comfortable headphones are suitable for listening to music, online classes, and everyday use. Their simple design makes them an easy choice

Task 2 - 1280px - top of the product page



SHOP EASY COLLECTION

Headphones

Rs. 2,500

Quantity

1

Add to Cart

Product Description

These comfortable headphones are suitable for listening to music, online classes, and everyday use. Their simple design makes them an easy choice for studying or relaxing.

The soft ear cushions sit gently around the ear, so you can wear them for a long study session without feeling pressure. The headband is adjustable and flexible, which means it fits small and large heads equally well. Because the frame is light, you will hardly notice it on your head even after a few hours.

Key Features

- Comfortable over-ear design
- Suitable for daily listening
- Simple and lightweight style
- Soft cushions for long listening sessions

Task 2 - 1280px - scrolled down: details box is sticky, 104px from the top

Code added / changed

product.html

```
<!-- TASK 2 START: CSS Grid layout with named areas (image / details / description) -->
        <section class="product-detail product-layout">
            <div class="product-image-box">
                <img src="headphones_img.jpg"
                     alt="Red headphones" class="detail-image">
            </div>

            <!-- This is the sticky box (position: sticky is set in style.css) -->
            <div class="product-details-box">
                <p class="small-label">SHOP EASY COLLECTION</p>
                <h1 class="detail-title">Headphones</h1>
                <p class="detail-price">Rs. 2,500</p>

                <div class="quantity-area">
                    <label for="quantity" class="form-label">Quantity</label>
                    <input type="number" id="quantity" name="quantity"
                           class="form-control quantity-input" min="1" value="1">
                </div>

                <button type="button" class="btn shop-button detail-cart-button">
                    Add to Cart
                </button>
            </div>

            <div class="product-description-box">
                <h2 class="description-heading">Product Description</h2>
                <p class="detail-description">
                    These comfortable headphones are suitable for listening to music,
                    online classes, and everyday use. Their simple design makes them
                    an easy choice for studying or relaxing.
                </p>
                <p class="detail-description">
                    The soft ear cushions sit gently around the ear, so you can wear them for
                    a long study session without feeling pressure. The headband is adjustable
                    and flexible, which means it fits small and large heads equally well. Because
                    the frame is light, you will hardly notice it on your head even after a few hours.
                </p>

                <h3 class="description-subheading">Key Features</h3>
                <ul class="product-features">
                    <li>Comfortable over-ear design</li>
                    <li>Suitable for daily listening</li>
                    <li>Simple and lightweight style</li>
                    <li>Soft cushions for long listening sessions</li>
                    <li>Adjustable headband that fits most people</li>
                    <li>Works with phones, laptops and tablets</li>
                </ul>

                <h3 class="description-subheading">Great For</h3>
                <p class="detail-description">
                    Online classes and video lectures, listening to music while travelling to
                    university, watching movies at home, and quiet study time in the library.
                    The sound is clear and balanced, so voices in lectures are easy to understand
                    and music still sounds pleasant.
                </p>

                <h3 class="description-subheading">Sound and Comfort</h3>
                <p class="detail-description">
                    The speakers inside each ear cup give clear voices and steady bass. You do not
                    need to turn the volume very high, which is better for your ears. The cushions
                    block some outside noise, so it is easier to focus when you are studying in
                    a busy place.
                </p>
                <p class="detail-description">
                    When you are not using them, the ear cups fold inwards, so the headphones fit
                    easily inside a backpack or travel bag. The cable is long enough for comfortable
                    use at a desk and does not tangle easily.
                </p>

                <h3 class="description-subheading">What Is in the Box</h3>
                <ul class="product-features">
                    <li>1 pair of headphones</li>
                    <li>1 audio cable</li>
                    <li>1 simple user guide</li>
                </ul>

                <h3 class="description-subheading">Care Instructions</h3>
                <p class="detail-description">
```

```
                    Wipe the cushions with a dry, soft cloth. Do not use water or strong cleaners.
                    Keep the headphones in a bag or case when travelling, and do not wrap the cable
                    too tightly around the headband. With simple care, they will last for a long time.
                </p>

                <h3 class="description-subheading">Delivery and Returns</h3>
                <p class="detail-description">
                    Orders are normally delivered within three to five working days. If the product
                    has a problem, you can ask for a replacement within seven days of delivery.
                    Please keep the box and the user guide until you are sure you are happy with
                    the headphones.
                </p>

                <p class="small text-muted mt-3">This is a static demo. The cart and quantity are not connected to a
backend.</p>
                <a href="index.html" class="back-link">← Back to all products</a>
            </div>
        </section>
        <!-- TASK 2 END -->
```

style.css

```
/* ===== TASK 2 START: product page with CSS Grid (named areas) ===== */
/* Below 674px: one column, stacked in this order: image, details, description */
.product-layout {
    display: grid;
    grid-template-columns: 1fr;
    grid-template-areas:
        "image"
        "details"
        "description";
    gap: 24px;
}

.product-image-box { grid-area: image; }
.product-details-box { grid-area: details; }
.product-description-box { grid-area: description; }

.product-details-box {
    background: #f8f9fa;
    border: 1px solid #dde2e7;
    border-top: 4px solid hsl(158, 70%, 45%);
    border-radius: 10px;
    padding: 20px;
}

.description-heading {
    color: var(--shop-blue);
    font-size: 26px;
    font-weight: bold;
    margin-bottom: 12px;
}

.description-subheading {
    color: var(--shop-blue);
    font-size: 20px;
    font-weight: bold;
    margin: 22px 0 8px;
}

/* From 674px and wider: image + details side by side, description full width underneath */
@media (min-width: 674px) {
    .product-layout {
        grid-template-columns: 1fr 1fr;
        grid-template-areas:
            "image       details"
            "description description";
        gap: 32px;
    }

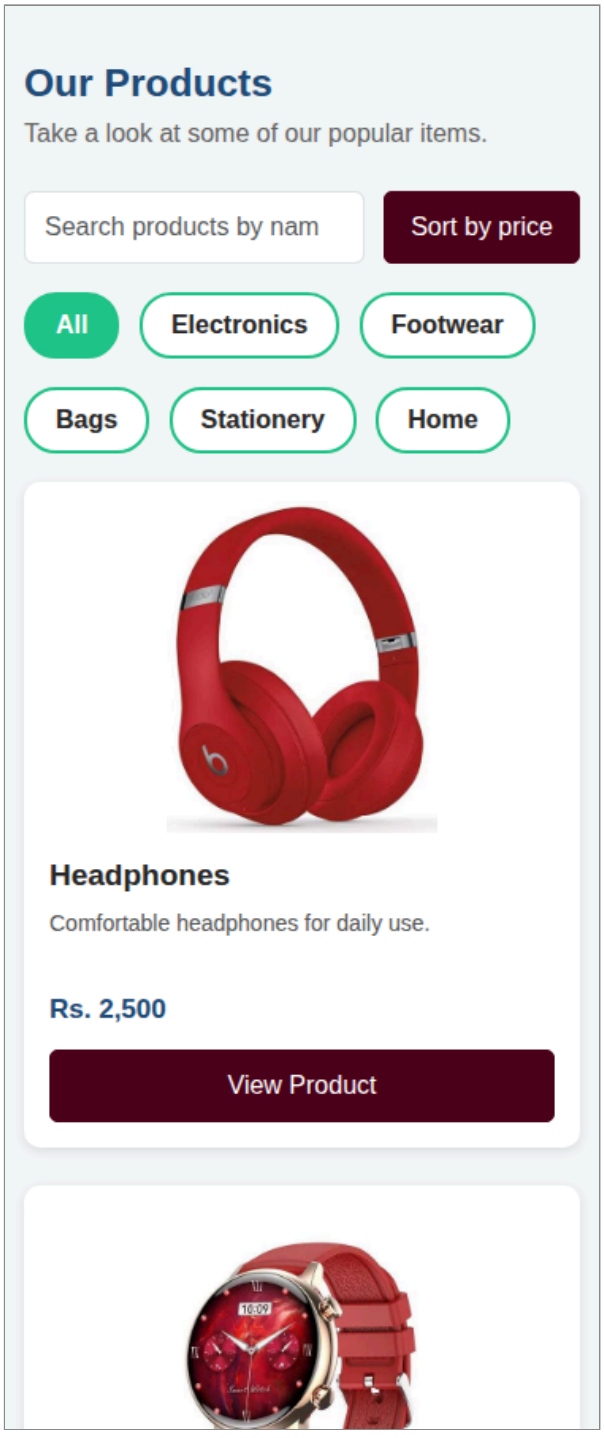
    /* tall image row: gives the sticky box room to move while you scroll */
    .product-image-box .detail-image {
        height: 640px;
        max-height: none;
    }

    /* sticky details box: stays 104px from the top of the screen  (74 mod 40 + 70 = 104) */
    .product-details-box {
        position: sticky;
        top: 104px;
        align-self: start;   /* box keeps its own height, so it can move inside the row */
    }
```

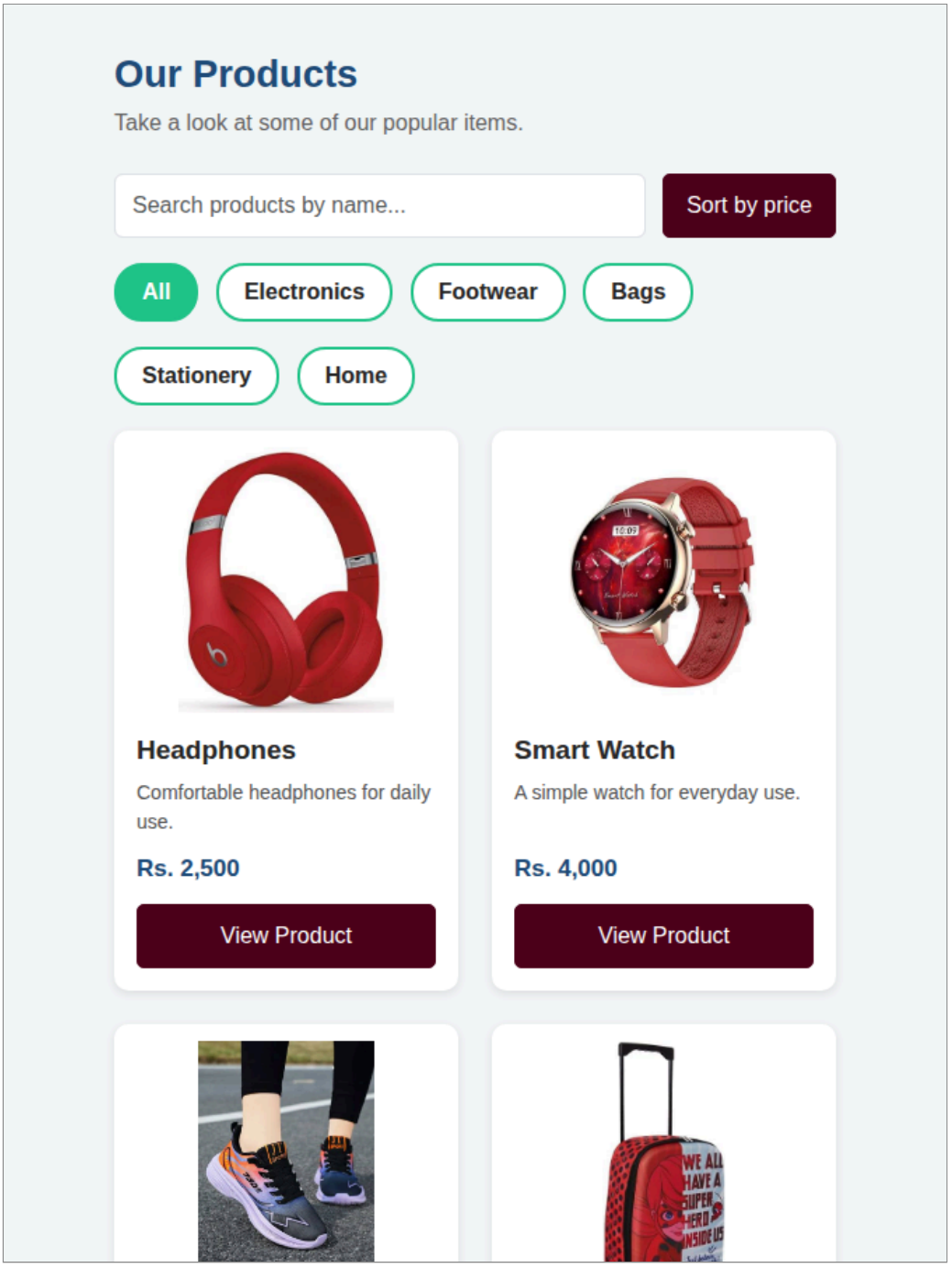
```
}
/* ===== TASK 2 END ===== */
```

Task 3 - JavaScript: live search and price sort

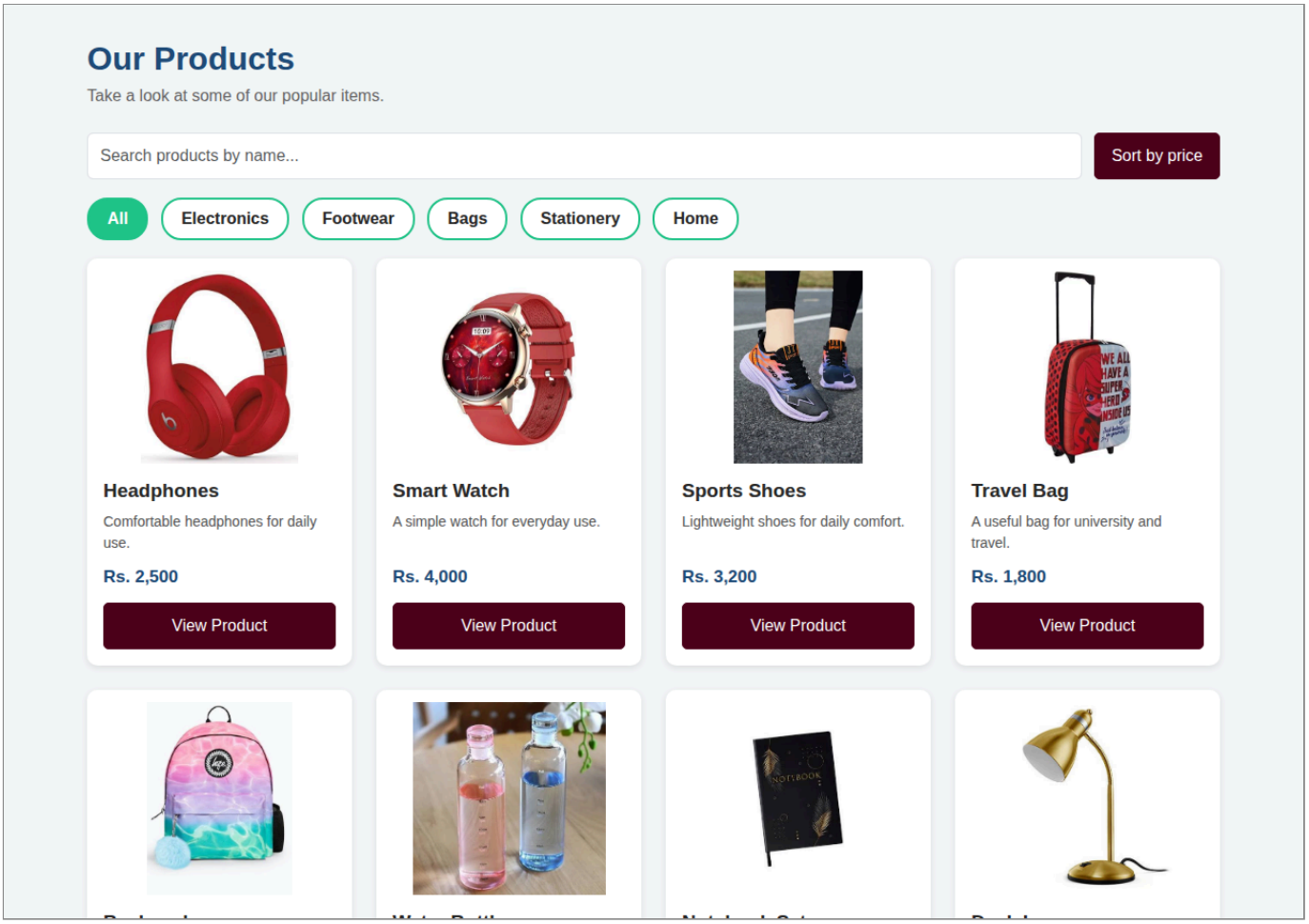
Screenshots (375px, 674px, 1280px, and after search / sort)



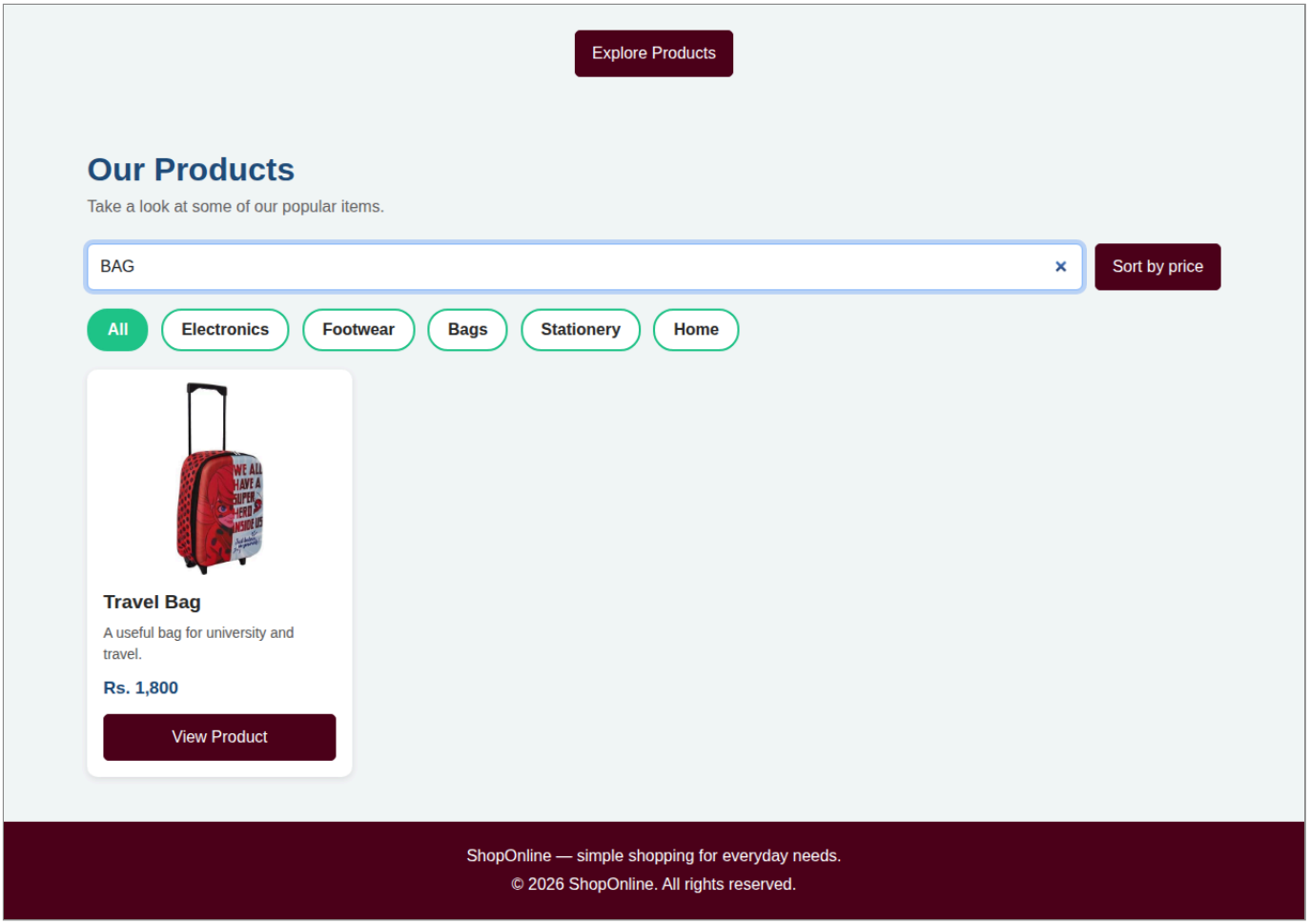
Task 3 - 375px - search box, sort button and cards



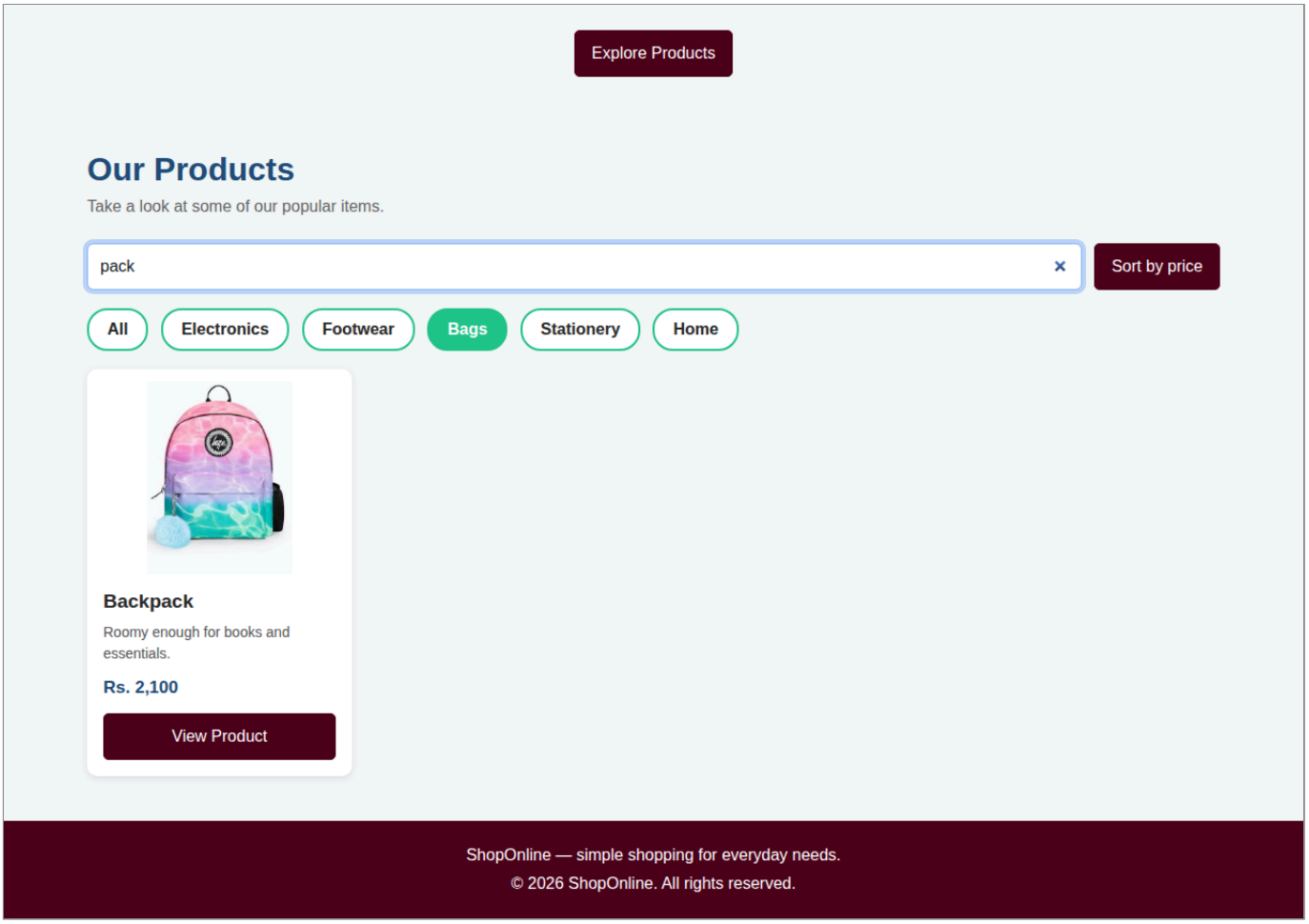
Task 3 - 674px - search box, sort button and cards



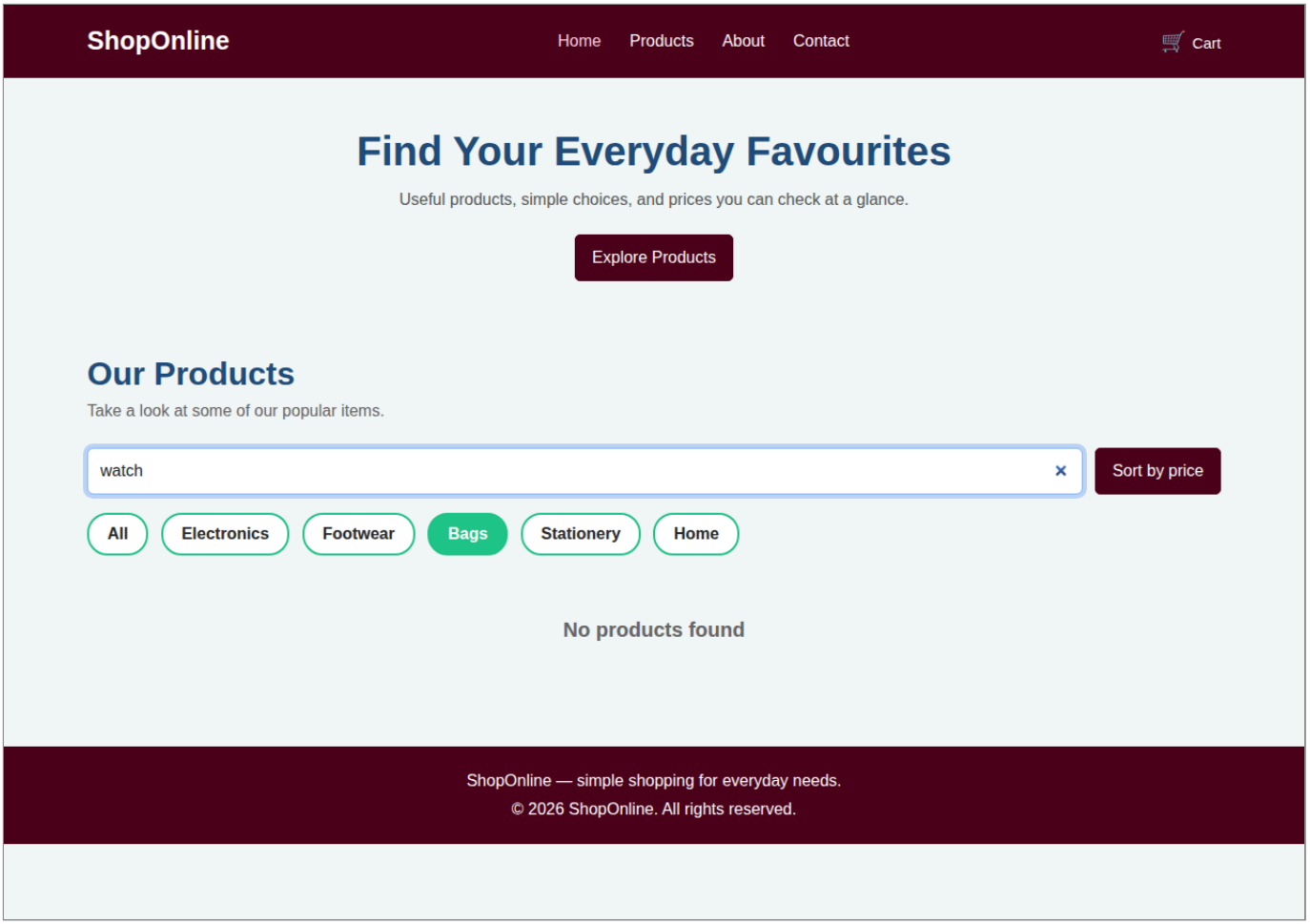
Task 3 - 1280px - search box, sort button and cards



Task 3 - 1280px - typed "BAG" (not case-sensitive): only Travel Bag stays



Task 3 - 1280px - category "Bags" + search "pack" together: only Backpack



Task 3 - 1280px - category "Bags" + search "watch": nothing matches, message shown

Our Products

Take a look at some of our popular items.

Search products by name...

Sort by price (Low to High)

All | Electronics | Footwear | Bags | Stationery | Home



Notebook Set

Simple notebooks for classes and notes.

Rs. 500

View Product



Water Bottle

A reusable bottle for your daily routine.

Rs. 750

View Product



Desk Lamp

A handy light for studying at night.

Rs. 1,400

View Product



Travel Bag

A useful bag for university and travel.

Rs. 1,800

View Product









Task 3 - 1280px - 1st click on Sort by price: lowest to highest

Our Products

Take a look at some of our popular items.

Search products by name...

Sort by price (High to Low)

All | Electronics | Footwear | Bags | Stationery | Home



Smart Watch

A simple watch for everyday use.

Rs. 4,000

View Product



Sports Shoes

Lightweight shoes for daily comfort.

Rs. 3,200

View Product



Headphones

Comfortable headphones for daily use.

Rs. 2,500

View Product



Backpack

Roomy enough for books and essentials.

Rs. 2,100

View Product









Task 3 - 1280px - 2nd click: highest to lowest

Our Products

Take a look at some of our popular items.

watch ×

Sort by price

All | Electronics | Footwear | Bags | Stationery | Home



Smart Watch

A simple watch for everyday use.

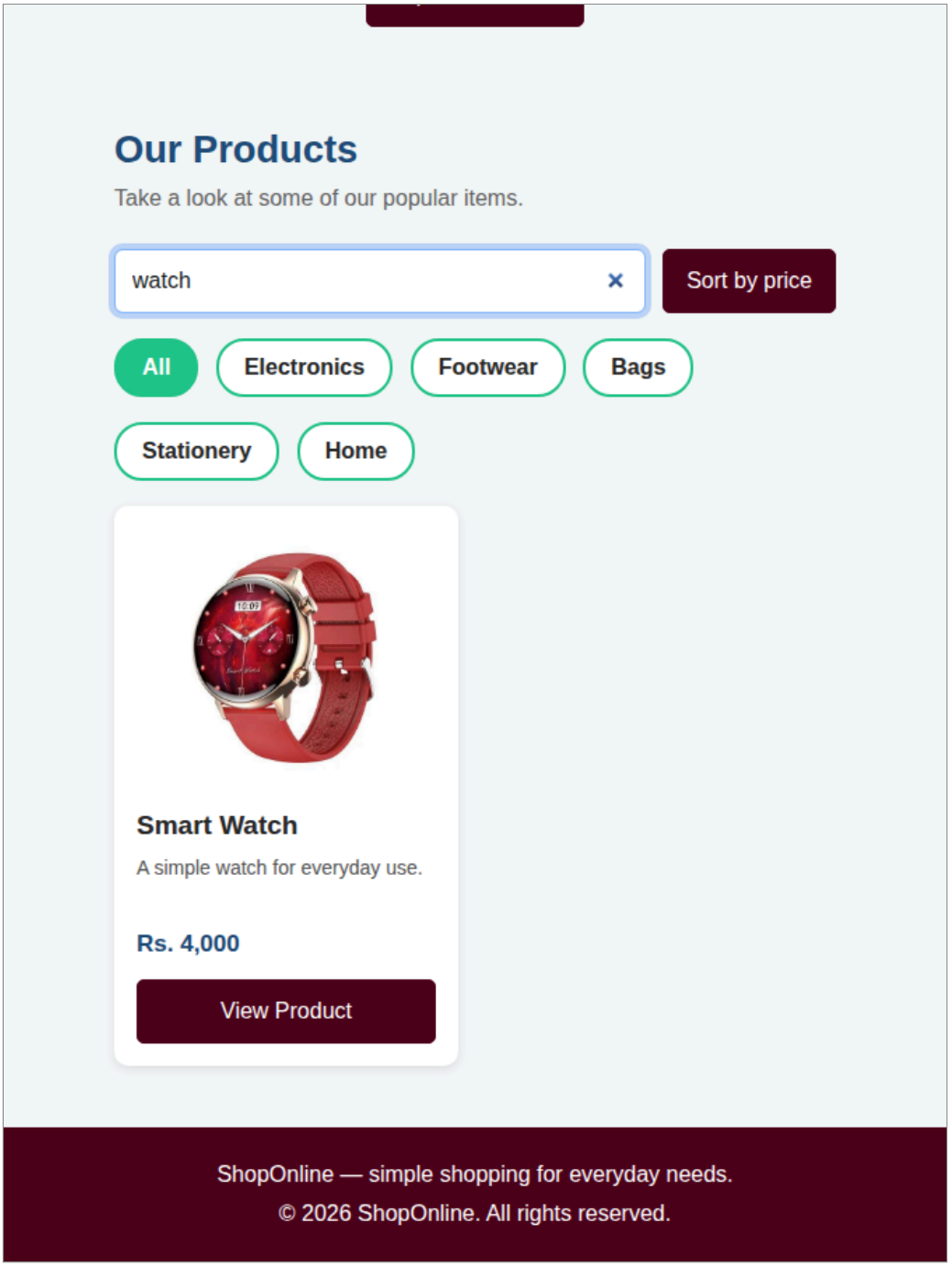
Rs. 4,000

View Product

ShopOnline — simple shopping for everyday needs.

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Task 3 - 375px - typed "watch": only Smart Watch stays



Task 3 - 674px - typed "watch": only Smart Watch stays

Code added / changed

index.html

```
<!-- TASK 3 START: search box + sort button -->
            <div class="shop-toolbar">
                <input type="search" id="search-box" class="form-control" placeholder="Search products by name..." aria-label="Search products">
                <button type="button" id="sort-btn" class="btn shop-button">Sort by price</button>
            </div>
            <!-- TASK 3 END -->

<!-- TASK 3: message shown by script.js when no card matches -->
                <p id="no-results" class="d-none">No products found</p>

<!-- at the end of index.html, after the bootstrap script: -->
<script src="script.js"></script>
```

style.css

```
/* ===== TASK 3 START: search box, sort button, Premium label ===== */
.shop-toolbar {
```

```
    display: flex;
    gap: 12px;
    margin-bottom: 18px;
}

.shop-toolbar #search-box {
    flex: 1;
}

.shop-toolbar #sort-btn {
    white-space: nowrap;
}

/* class added by script.js to cards that do not match the search text */
.search-hidden {
    display: none !important;
}

#no-results {
    text-align: center;
    font-size: 20px;
    font-weight: bold;
    color: #666;
    padding: 40px 0;
}

/* "Premium" label added by script.js */
.product-card {
    position: relative;
}

.premium-label {
    position: absolute;
    top: 10px;
    left: 10px;
    background: hsl(158, 70%, 45%);
    color: white;
    font-size: 12px;
    font-weight: bold;
    padding: 3px 10px;
    border-radius: 12px;
}
/* ===== TASK 3 END ===== */
```

script.js (new file)

```
// Task 3 - live search, price sort and Premium label (plain JavaScript)

const N = 74;                          // last two digits of my roll number
const PREMIUM_LIMIT = N * 100;         // products above this price get a Premium label

const grid      = document.getElementById("product-grid");
const searchBox = document.getElementById("search-box");
const sortBtn   = document.getElementById("sort-btn");
const message   = document.getElementById("no-results");

// the existing cards (the columns that are already in index.html)
const cards = grid.querySelectorAll(".product-item");

// read the price from the card itself: "Rs. 2,500" -> 2500
function getPrice(card) {
    const text = card.querySelector(".price").textContent;
    return Number(text.replace(/[^0-9]/g, ""));
}

// ---------- 1. Premium label (runs when the page loads) ----------
cards.forEach(function (card) {
    if (getPrice(card) > PREMIUM_LIMIT) {
        const label = document.createElement("span");
        label.className = "premium-label";
        label.textContent = "Premium";
        card.querySelector(".product-card").appendChild(label);
    }
});

// ---------- 2. Search + category together ----------
function applyFilters() {
    const text = searchBox.value.trim().toLowerCase();
    const category = document.querySelector('input[name="category"]:checked').value;
    let visibleCount = 0;

    cards.forEach(function (card) {
```

```
        const name = card.querySelector("h3").textContent.toLowerCase();
        const matchesSearch = name.includes(text);
        const matchesCategory = category === "all" || card.dataset.category === category;

        // the search hides cards with a class; the category radio hides cards with CSS (Task 1)
        card.classList.toggle("search-hidden", !matchesSearch);

        // a card is really visible only if it matches BOTH
        if (matchesSearch && matchesCategory) {
            visibleCount++;
        }
    });

    // nothing visible -> show the message in place of the grid
    message.classList.toggle("d-none", visibleCount > 0);
    grid.classList.toggle("d-none", visibleCount === 0);
}

searchBox.addEventListener("input", applyFilters);

// also re-check when the user changes the category radio buttons
document.querySelectorAll('input[name="category"]').forEach(function (radio) {
    radio.addEventListener("change", applyFilters);
});

// ---------- 3. Sort by price (low->high, then high->low, ...) ----------
let ascending = true;

sortBtn.addEventListener("click", function () {
    const list = Array.from(grid.children);          // the existing card elements

    list.sort(function (a, b) {
        return ascending ? getPrice(a) - getPrice(b) : getPrice(b) - getPrice(a);
    });

    // appendChild MOVES each existing element, so the grid is reordered (no innerHTML)
    list.forEach(function (card) {
        grid.appendChild(card);
    });

    sortBtn.textContent = ascending ? "Sort by price (Low to High)" : "Sort by price (High to Low)";
    ascending = !ascending;                          // next click goes the other way
});
```